Requirements to Tests Map

CSE 360, Fall 2026 (C) · Team 24: Arush Ugal, Karan Patel, Anirudh B, Pranjal Ganvir, Moksha Dev
First issued October 22, 2026; this revision October 26, 2026

What this map covers

CS-1 to CS-12 are the Contributor user stories split into requirements that can each be tested on their own (agreed by the team on October 13). Every requirement has at least one positive test, showing it is met for valid use, and at least one negative test, showing incorrect use is refused or explained. There are 83 tests: 76 semi-automated tests in four suites, run together by `TP2TestRunner`, and 7 manual procedures (MAN) for what only a window can show. Every test is one method; its Javadoc states the requirement, how the test works, and what output means a pass. The HTML pages and screenshots named here are in the `ContributorTests_Javadocs` folder.

How to run the tests

Automated	Run `testingAutomation.TP2TestRunner` (Eclipse: Run As, Java Application, with the JavaFX module path and the H2 jar as for `FoundationsMain`). Each test prints one line starting `Pass:` or `Failure:` followed by its id and requirement; the last line is `RESULT: every test behaved as expected` only when nothing failed.
Manual	Run `testingAutomation.LessonsManualTestProcedures` beside a running `FoundationsMain`. It prints each procedure's steps and expected result and records P, F, or S for each.

Requirements and their tests

Req.	Requirement	Positive tests	Negative tests	HTML pages
CS-1	A contributor can create a lesson with a title, category, description, and optional resolution and keywords. The contributor is set from the logged-in user.	CRUD-01, MOD-01, MAN-02	CRUD-02, MOD-02	LessonsCrudTestingAutomation.html LessonsManualTestProcedures.html LessonsModelTestingAutomation.html
CS-2	A contributor can see all lessons, newest first. An empty list is explained in words, not shown as a blank screen.	LIST-01, LIST-11, MOD-03	LIST-02, MOD-19, MAN-01	LessonsListTestingAutomation.html LessonsManualTestProcedures.html LessonsModelTestingAutomation.html
CS-3	A contributor can open one lesson and see all of its fields.	CRUD-03	CRUD-04	LessonsCrudTestingAutomation.html
CS-4	A contributor can search lessons. The search covers title, description, and keywords and ignores case. No match is explained in words.	LIST-03, LIST-10, LIST-12, MOD-05	LIST-04, LIST-13, MOD-06, MOD-17	LessonsListTestingAutomation.html LessonsModelTestingAutomation.html
CS-5	A contributor can filter lessons by category. An empty category is explained in words. The filter's label "All categories" cannot be used as a category.	LIST-05, LIST-09, LIST-10, MOD-07, MOD-24	LIST-06, MOD-08, MOD-18, MOD-23	LessonsListTestingAutomation.html LessonsModelTestingAutomation.html
CS-6	A contributor can view only their own lessons. If they have none, a message says so.	LIST-07, MOD-09	LIST-08, MOD-04, MOD-10	LessonsListTestingAutomation.html LessonsModelTestingAutomation.html

Req.	Requirement	Positive tests	Negative tests	HTML pages
CS-7	A contributor can update a lesson they contributed. An update is validated the same way as a new lesson.	CRUD-05, MOD-13	CRUD-06, MOD-14	LessonsCrudTestingAutomation.html LessonsModelTestingAutomation.html
CS-8	A contributor can delete a lesson they contributed, after confirming. It is removed for all users.	CRUD-07, MOD-15, MAN-04, MAN-06	CRUD-08, MOD-16, MAN-04	LessonsCrudTestingAutomation.html LessonsManualTestProcedures.html LessonsModelTestingAutomation.html
CS-9	Only the contributor of a lesson can update or delete it. Any other user is refused with a message.	CRUD-09, MOD-11	CRUD-10, MOD-12, MOD-16, MOD-20, MAN-05	LessonsCrudTestingAutomation.html LessonsManualTestProcedures.html LessonsModelTestingAutomation.html
CS-10	Title, category, description, and contributor are required; resolution and keywords are optional. A missing required value is refused with a message naming the field.	VAL-01, VAL-06, VAL-10, VAL-15, VAL-18, VAL-22, VAL-24	VAL-02, VAL-03, VAL-07, VAL-11, VAL-23, VAL-25, MAN-03	LessonValidationTestingAutomation.html LessonsManualTestProcedures.html
CS-11	Length limits: title 80, category 30, description 2000, resolution 2000, keywords 200 characters, at most 10 keywords of 30 characters each. A value over a limit, or an empty keyword entry, is refused with a message naming the problem.	VAL-04, VAL-08, VAL-12, VAL-13, VAL-16, VAL-19, VAL-24	VAL-05, VAL-09, VAL-14, VAL-17, VAL-20, VAL-21, MOD-21, MOD-22	LessonValidationTestingAutomation.html LessonsModelTestingAutomation.html
CS-12	Lessons persist across application restarts, including on a database left by an HW2 version.	CRUD-11, CRUD-13, MAN-07	CRUD-12, CRUD-14	LessonsCrudTestingAutomation.html LessonsManualTestProcedures.html

Every test, its Javadoc, and its screenshot

The anchor is the method name inside the HTML page: `#crud01CreateValidLesson()` in `LessonsCrudTestingAutomation.html` opens that test's Javadoc. CRUD-11 and CRUD-12, and CRUD-13 and CRUD-14, share one method because the negative test checks the database the positive test reopened.

Test	Req.	Kind	Method (anchor in the HTML page)	Screenshot
LessonValidationTestingAutomation.html				
VAL-01	CS-10	pos.	`#val01TitleAccepted()`	`7.4-VAL-01-TitleAccepted.png`
VAL-02	CS-10	neg.	`#val02BlankTitleRefused()`	`7.4-VAL-02-BlankTitleRefused.png`
VAL-03	CS-10	neg.	`#val03WhitespaceTitleRefused()`	`7.4-VAL-03-WhitespaceTitleRefused.png`
VAL-04	CS-11	pos.	`#val04TitleAt80Accepted()`	`7.4-VAL-04-TitleAt80Accepted.png`
VAL-05	CS-11	neg.	`#val05TitleAt81Refused()`	`7.4-VAL-05-TitleAt81Refused.png`
VAL-06	CS-10	pos.	`#val06CategoryAccepted()`	`7.4-VAL-06-CategoryAccepted.png`
VAL-07	CS-10	neg.	`#val07BlankCategoryRefused()`	`7.4-VAL-07-BlankCategoryRefused.png`
VAL-08	CS-11	pos.	`#val08CategoryAt30Accepted()`	`7.4-VAL-08-CategoryAt30Accepted.png`
VAL-09	CS-11	neg.	`#val09CategoryAt31Refused()`	`7.4-VAL-09-CategoryAt31Refused.png`
VAL-10	CS-10	pos.	`#val10DescriptionAccepted()`	`7.4-VAL-10-DescriptionAccepted.png`
VAL-11	CS-10	neg.	`#val11BlankDescriptionRefused()`	`7.4-VAL-11-BlankDescriptionRefused.png`
VAL-12	CS-11	pos.	`#val12DescriptionAt300Accepted()`	`7.4-VAL-12-DescriptionAt300Accepted.png`
VAL-13	CS-11	pos.	`#val13DescriptionAt2000Accepted()`	`7.4-VAL-13-DescriptionAt2000Accepted.png`
VAL-14	CS-11	neg.	`#val14DescriptionAt2001Refused()`	`7.4-VAL-14-DescriptionAt2001Refused.png`

Test	Req.	Kind	Method (anchor in the HTML page)	Screenshot
VAL-15	CS-10	pos.	`#val15EmptyResolutionAccepted()`	`7.4-VAL-15-EmptyResolutionAccepted.png`
VAL-16	CS-11	pos.	`#val16ResolutionAt2000Accepted()`	`7.4-VAL-16-ResolutionAt2000Accepted.png`
VAL-17	CS-11	neg.	`#val17ResolutionAt2001Refused()`	`7.4-VAL-17-ResolutionAt2001Refused.png`
VAL-18	CS-10	pos.	`#val18EmptyKeywordsAccepted()`	`7.4-VAL-18-EmptyKeywordsAccepted.png`
VAL-19	CS-11	pos.	`#val19ValidKeywordsAccepted()`	`7.4-VAL-19-ValidKeywordsAccepted.png`
VAL-20	CS-11	neg.	`#val20ElevenKeywordsRefused()`	`7.4-VAL-20-ElevenKeywordsRefused.png`
VAL-21	CS-11	neg.	`#val21LongKeywordRefused()`	`7.4-VAL-21-LongKeywordRefused.png`
VAL-22	CS-10	pos.	`#val22ContributorAccepted()`	`7.4-VAL-22-ContributorAccepted.png`
VAL-23	CS-10	neg.	`#val23BlankContributorRefused()`	`7.4-VAL-23-BlankContributorRefused.png`
VAL-24	CS-10, CS-11	pos.	`#val24FullLessonAccepted()`	`7.4-VAL-24-FullLessonAccepted.png`
VAL-25	CS-10	neg.	`#val25FullLessonWithBlankTitleRefused()`	`7.4-VAL-25-FullLessonWithBlankTitleRefused.png`
LessonsCrudTestingAutomation.html				
CRUD-01	CS-1	pos.	`#crud01CreateValidLesson()`	`7.4-CRUD-01-CreateValidLesson.png`
CRUD-02	CS-1	neg.	`#crud02CreateBlankTitleRefused()`	`7.4-CRUD-02-CreateBlankTitleRefused.png`
CRUD-03	CS-3	pos.	`#crud03ReadById()`	`7.4-CRUD-03-ReadById.png`
CRUD-04	CS-3	neg.	`#crud04ReadUnknownId()`	`7.4-CRUD-04-ReadUnknownId.png`
CRUD-05	CS-7	pos.	`#crud05UpdatePersists()`	`7.4-CRUD-05-UpdatePersists.png`
CRUD-06	CS-7	neg.	`#crud06UpdateBlankTitleRefused()`	`7.4-CRUD-06-UpdateBlankTitleRefused.png`
CRUD-07	CS-8	pos.	`#crud07Delete()`	`7.4-CRUD-07-Delete.png`
CRUD-08	CS-8	neg.	`#crud08DeleteTwice()`	`7.4-CRUD-08-DeleteTwice.png`
CRUD-09	CS-9	pos.	`#crud09OwnerRecognised()`	`7.4-CRUD-09-OwnerRecognised.png`
CRUD-10	CS-9	neg.	`#crud10NonOwnerNotRecognised()`	`7.4-CRUD-10-NonOwnerNotRecognised.png`
CRUD-11	CS-12	pos.	`#crud11And12PersistAcrossReopen()`	`7.4-CRUD-11-CRUD-12-PersistAcrossReopen.png`
CRUD-12	CS-12	neg.	`#crud11And12PersistAcrossReopen()`	`7.4-CRUD-11-CRUD-12-PersistAcrossReopen.png`
CRUD-13	CS-12	pos.	`#crud13And14OpenHw2Layout()`	`7.4-CRUD-13-CRUD-14-OpenHw2Layout.png`
CRUD-14	CS-12	neg.	`#crud13And14OpenHw2Layout()`	`7.4-CRUD-13-CRUD-14-OpenHw2Layout.png`
LessonsListTestingAutomation.html				
LIST-01	CS-2	pos.	`#list01ReadAllReturnsEveryLesson()`	`7.4-LIST-01-ReadAllReturnsEveryLesson.png`
LIST-02	CS-2	neg.	`#list02EmptyDatabaseGivesEmptyList()`	`7.4-LIST-02-EmptyDatabaseGivesEmptyList.png`
LIST-03	CS-4	pos.	`#list03SearchFindsKeyword()`	`7.4-LIST-03-SearchFindsKeyword.png`
LIST-04	CS-4	neg.	`#list04SearchForAbsentTerm()`	`7.4-LIST-04-SearchForAbsentTerm.png`
LIST-05	CS-5	pos.	`#list05FilterByCategory()`	`7.4-LIST-05-FilterByCategory.png`
LIST-06	CS-5	neg.	`#list06FilterByAbsentCategory()`	`7.4-LIST-06-FilterByAbsentCategory.png`
LIST-07	CS-6	pos.	`#list07MyLessons()`	`7.4-LIST-07-MyLessons.png`
LIST-08	CS-6	neg.	`#list08LessonsOfUnknownUser()`	`7.4-LIST-08-LessonsOfUnknownUser.png`

Test	Req.	Kind	Method (anchor in the HTML page)	Screenshot
LIST-09	CS-5	pos.	`#list09CategoriesOffered()`	`7.4-LIST-09-CategoriesOffered.png`
LIST-10	CS-4, CS-5	pos.	`#list10SearchThenFilter()`	`7.4-LIST-10-SearchThenFilter.png`
LIST-11	CS-2	pos.	`#list11NewestFirst()`	`7.4-LIST-11-NewestFirst.png`
LIST-12	CS-4	pos.	`#list12SearchIgnoresCase()`	`7.4-LIST-12-SearchIgnoresCase.png`
LIST-13	CS-4	neg.	`#list13EmptySearchTerm()`	`7.4-LIST-13-EmptySearchTerm.png`
LessonsModelTestingAutomation.html				
MOD-01	CS-1	pos.	`#mod01SaveNewLesson()`	`7.4-MOD-01-SaveNewLesson.png`
MOD-02	CS-1	neg.	`#mod02SaveNewLessonBlankTitle()`	`7.4-MOD-02-SaveNewLessonBlankTitle.png`
MOD-03	CS-2	pos.	`#mod03NoFiltersShowsAll()`	`7.4-MOD-03-NoFiltersShowsAll.png`
MOD-04	CS-6	neg.	`#mod04MessageWhenUserHasNoLessons()`	`7.4-MOD-04-MessageWhenUserHasNoLessons.png`
MOD-05	CS-4	pos.	`#mod05Search()`	`7.4-MOD-05-Search.png`
MOD-06	CS-4	neg.	`#mod06SearchForAbsentTerm()`	`7.4-MOD-06-SearchForAbsentTerm.png`
MOD-07	CS-5	pos.	`#mod07CategoryFilter()`	`7.4-MOD-07-CategoryFilter.png`
MOD-08	CS-5	neg.	`#mod08AbsentCategory()`	`7.4-MOD-08-AbsentCategory.png`
MOD-09	CS-6	pos.	`#mod09OnlyMine()`	`7.4-MOD-09-OnlyMine.png`
MOD-10	CS-6	neg.	`#mod10OnlyMineForUserWithNone()`	`7.4-MOD-10-OnlyMineForUserWithNone.png`
MOD-11	CS-9	pos.	`#mod11OwnerMayChange()`	`7.4-MOD-11-OwnerMayChange.png`
MOD-12	CS-9	neg.	`#mod12NonOwnerMayNotChange()`	`7.4-MOD-12-NonOwnerMayNotChange.png`
MOD-13	CS-7	pos.	`#mod13OwnerUpdate()`	`7.4-MOD-13-OwnerUpdate.png`
MOD-14	CS-7	neg.	`#mod14UpdateBlankDescription()`	`7.4-MOD-14-UpdateBlankDescription.png`
MOD-15	CS-8	pos.	`#mod15OwnerDelete()`	`7.4-MOD-15-OwnerDelete.png`
MOD-16	CS-8, CS-9	neg.	`#mod16NonOwnerDelete()`	`7.4-MOD-16-NonOwnerDelete.png`
MOD-17	CS-4	neg.	`#mod17MessageForNoSearchMatch()`	`7.4-MOD-17-MessageForNoSearchMatch.png`
MOD-18	CS-5	neg.	`#mod18MessageForNoCategoryMatch()`	`7.4-MOD-18-MessageForNoCategoryMatch.png`
MOD-19	CS-2	neg.	`#mod19MessageWhenNoLessonsStored()`	`7.4-MOD-19-MessageWhenNoLessonsStored.png`
MOD-20	CS-9	neg.	`#mod20NonOwnerEdit()`	`7.4-MOD-20-NonOwnerEdit.png`
MOD-21	CS-11	neg.	`#mod21LeadingEmptyKeywordRefused()`	`7.4-MOD-21-LeadingEmptyKeywordRefused.png`
MOD-22	CS-11	neg.	`#mod22TrailingEmptyKeywordRefused()`	`7.4-MOD-22-TrailingEmptyKeywordRefused.png`
MOD-23	CS-5	neg.	`#mod23ReservedCategoryRefused()`	`7.4-MOD-23-ReservedCategoryRefused.png`
MOD-24	CS-5	pos.	`#mod24SimilarCategoryNamesAccepted()`	`7.4-MOD-24-SimilarCategoryNamesAccepted.png`
LessonsManualTestProcedures.html				
MAN-01	CS-2	neg.	`#man01EmptyListMessage()`	`7.4-MAN-01-EmptyListMessage.png`
MAN-02	CS-1	pos.	`#man02CreateSetsContributor()`	`7.4-MAN-02-CreateSetsContributor.png`
MAN-03	CS-10	neg.	`#man03RefusedFieldIsNamed()`	`7.4-MAN-03-RefusedFieldIsNamed.png`
MAN-04	CS-8	pos. + neg.	`#man04DeleteAsksFirst()`	`7.4-MAN-04-DeleteAsksFirst.png`
MAN-05	CS-9	neg.	`#man05OtherUserCannotChange()`	`7.4-MAN-05-OtherUserCannotChange.png`
MAN-06	CS-8	pos.	`#man06DeleteSeenByOtherUser()`	`7.4-MAN-06-DeleteSeenByOtherUser.png`
MAN-07	CS-12	pos.	`#man07RestartKeepsLessons()`	`7.4-MAN-07-RestartKeepsLessons.png`

Other screenshots in ContributorTests_Javadocs

Screenshot	Shows
`7.4-<suite>-class-javadoc.png` (five)	Each suite's class Javadoc: requirements covered, test data, how to read the output
`7.4-TP2TestRunner-class-javadoc.png`, `-main.png`	What each suite covers; how the totals are added
`7.4-check-helper-and-refusedWith.png`	How a Pass or Failure line is produced; why a negative test checks the refusal message
`7.4-TP2TestRunner-output-1` to `-5`	The console output of a real run, one image per suite and one for the totals

Latest run

October 26, 2026, on the submitted source:

Suite	Tests	Passed	Failed
LessonValidationTestingAutomation (VAL)	25	25	0
LessonsCrudTestingAutomation (CRUD)	14	14	0
LessonsListTestingAutomation (LIST)	13	13	0
LessonsModelTestingAutomation (MOD)	24	24	0
Total, automated	**76**	**76**	**0**

To check that the tests can fail, three defects were put into a copy of the code: the HW2-table fix removed, keywords dropped from the search, and the list order reversed. The run then reported six failures (CRUD-13, CRUD-14, LIST-03, LIST-11, LIST-12, MOD-05) and no false passes.

The manual procedures are carried out during the TestingScreencast; their results are recorded there.

Iteration notes (Task 6)

Date	What the tests found, and what changed
Oct 14	VAL-12 (a 300-character description) failed on its first run. `InputLimits.checkLength` stops at 256 characters, so every description from 257 to 2000 was refused. Added `LessonInputPolicy.checkLongText` for description and resolution.
Oct 19	First CRUD run: 10 of 12. CRUD-02 and CRUD-06 failed because `create` and `update` wrote the row without acting on `validate`'s result. Both now return the problem and write nothing.
Oct 22	First GUI Model run: MOD-04 failed. `statusMessage` said nothing matched when only "Only my lessons" was ticked; the branch for that case now comes first. Full run: 63 of 63.
Oct 25	Review before submission. (1) Saving over the `lessonsLearned` table left by an HW2 version (columns author, body, tags) failed with "Column DESCRIPTION not found", and the list showed nothing. `LessonsLearnedDatabase` now renames such a table to `lessonsLearnedHW2Legacy`, keeping its rows; CRUD-13 and CRUD-14 added. (2) CRUD-11 and CRUD-12 only counted rows in an in-memory database, which cannot show persistence; they now close and reopen a database file. (3) Each test became its own documented method; negative validation tests now check the refusal message. (4) Added LIST-11 to LIST-13 (order, case, empty term), MOD-17 to MOD-20 (no-match messages, editing someone else's lesson), and MAN-01 to MAN-07. Result: 72 of 72.
Oct 26	Moksha's review of the editor (pull request #1) found two more defects. An empty first keyword (", java") was silently removed by `tidyKeywords` and the lesson accepted, and `validateKeywords` dropped a trailing empty entry ("java,"). A lesson could also be stored under "All categories", the filter's own label, and then never be shown on its own. Fixed in `ModelLessonEditor` and `LessonInputPolicy`, with the label now a single constant shared by the filter and the rule; MOD-21 to MOD-24 added. Result: 76 of 76.